

NMDS Ordination Analysis Report

Ördin v3.0 - Community Ecology Analysis Platform

October 25, 2025

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1 Executive Summary

This report presents the results of **Non-metric Multidimensional Scaling (NMDS)** ordination analysis performed on Dune Meadow Vegetation. The analysis reduces the complexity of multivariate community data into 2 dimensions while preserving rank-order dissimilarities between samples.

Key Findings:

- **Stress Value:** 0.1183 (B quality)
- **Convergence:** Successful
- **Distance Metric:** BRAY dissimilarity
- **Interpretation:** Fair representation (stress < 0.20)

Recommendation: Consider: (1) increasing dimensions (k), (2) trying different distance metric, or (3) checking for outliers.

2 Analysis Overview

Analysis Type: Non-metric Multidimensional Scaling (NMDS)

Dataset: Dune Meadow Vegetation

Number of Sites: 20

Number of Species: 30

Distance Metric: BRAY

Dimensions (k): 2

3 Ordination Quality Assessment

Stress: 0.1183 [Grade: B]

Fair representation (stress < 0.20)

The configuration should be used with caution. Some distortion of original distances is present.

Recommendation: Consider: (1) increasing dimensions (k), (2) trying different distance metric, or (3) checking for outliers.

Convergence: Solution converged

Iterations: 58

Non-metric R²: 0.9860

4 NMDS Ordination Plot

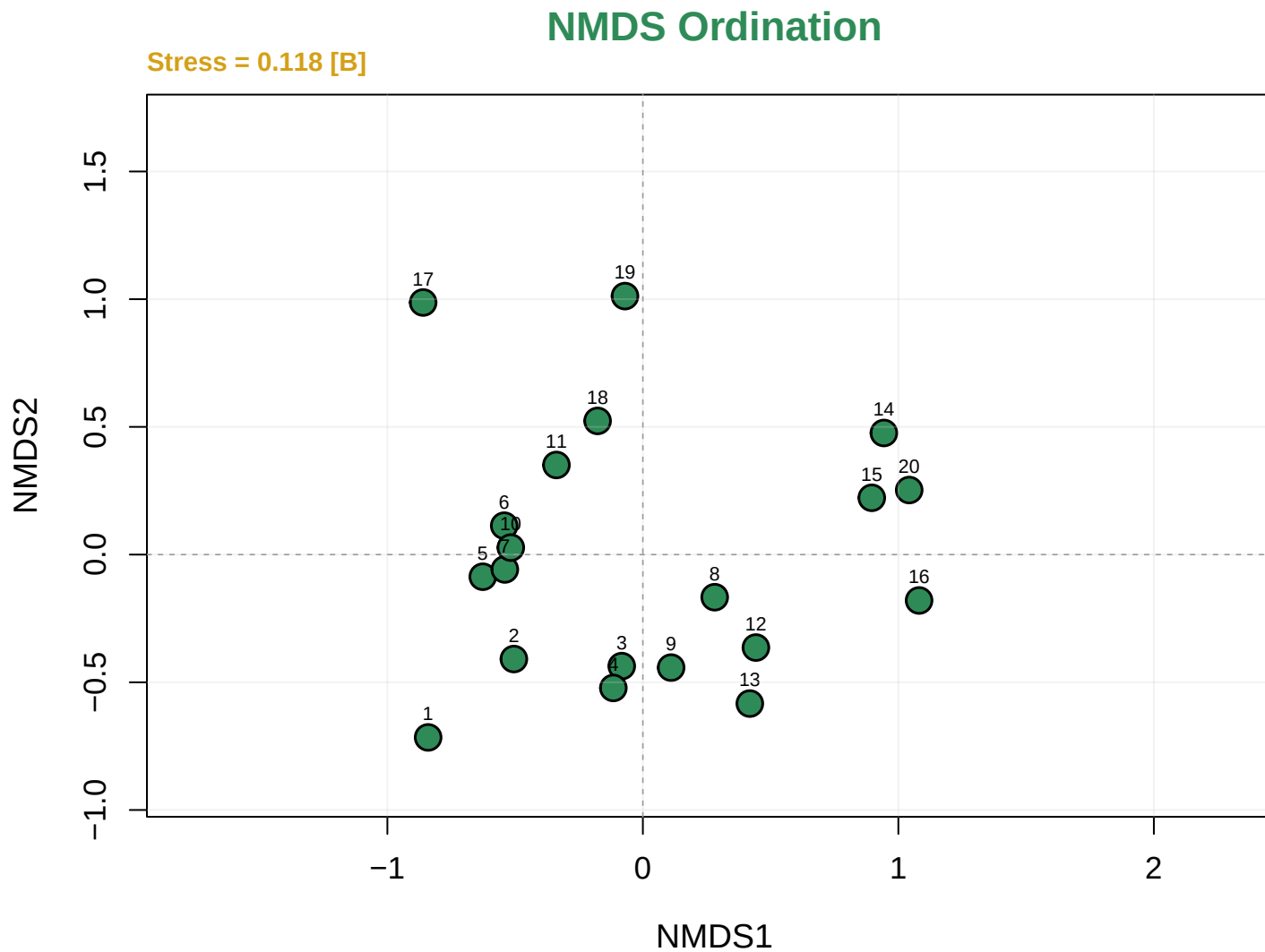


Figure 1: NMDS ordination showing sample positions in reduced dimensional space. Points represent individual samples, with distances approximately preserving ranked dissimilarities.

5 Diagnostic Plots

5.1 Shepard Plot

The Shepard plot compares observed dissimilarities with ordination distances. Points closer to the line indicate better fit.

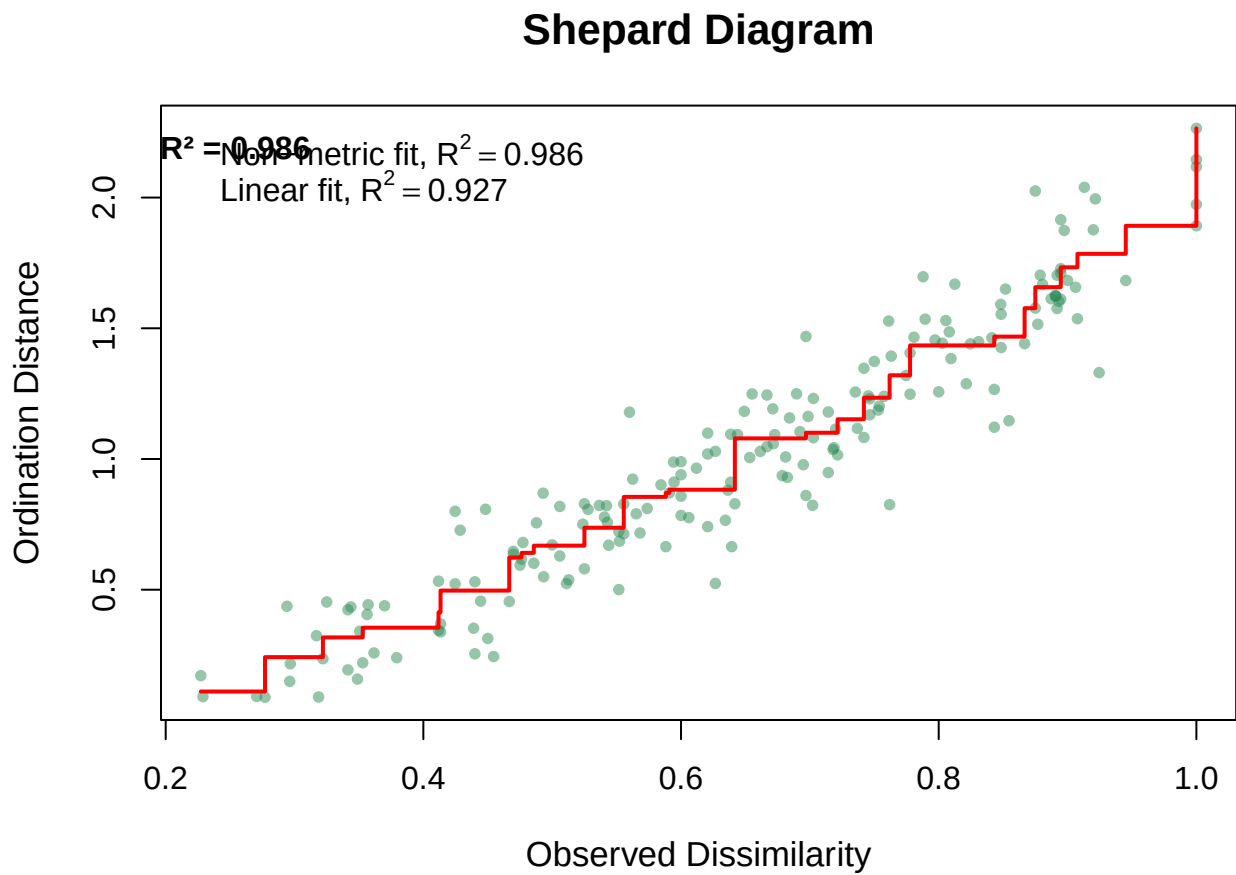


Figure 2: Shepard diagram showing the relationship between observed dissimilarities and ordination distances. The step line shows monotonic relationship, while points show actual fit.

5.2 Goodness of Fit by Site

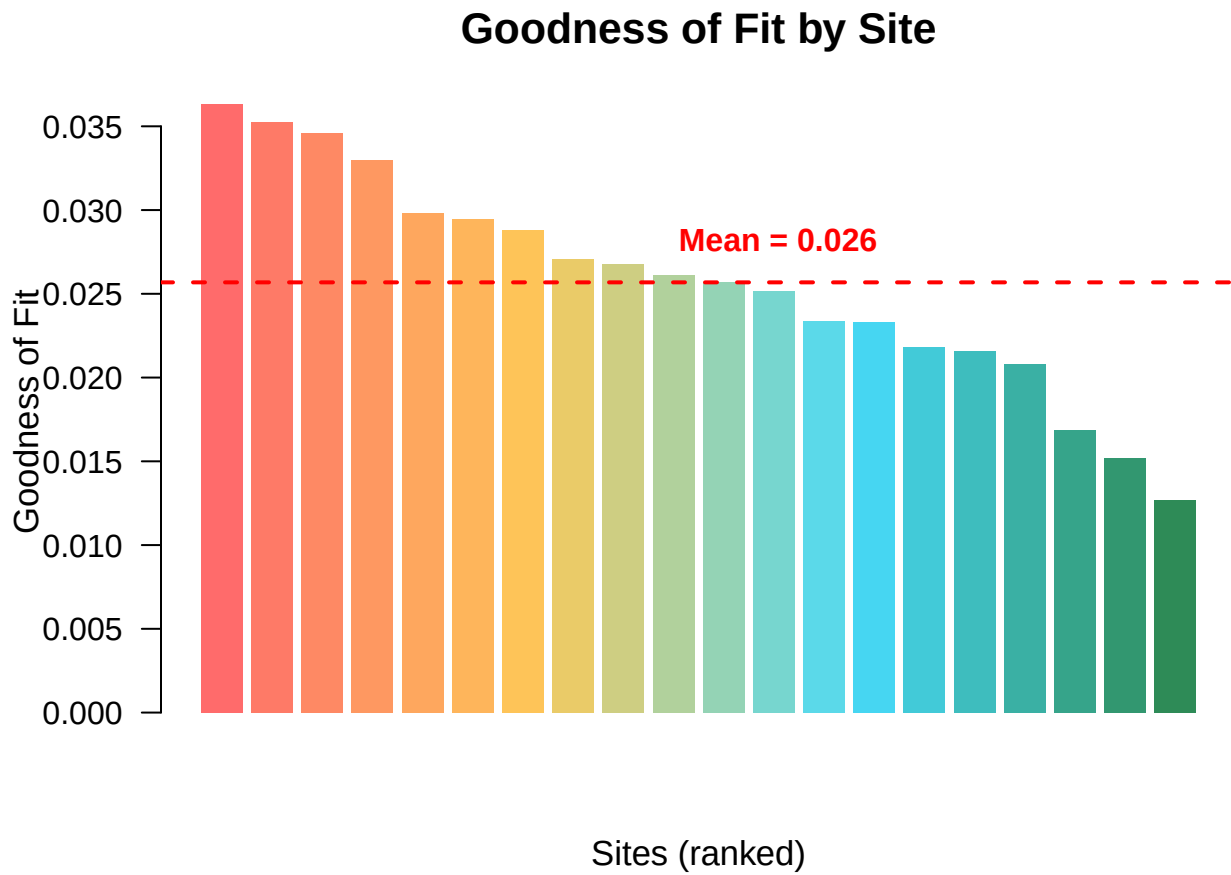


Figure 3: Goodness of fit for each sample site. Higher values indicate better representation in the ordination space.

6 Ordination Statistics

Table 1: NMDS Analysis Statistics

Statistic	Value
Stress	0.1183
Quality Grade	B
Convergence	Converged
Dimensions	2
Distance Metric	BRAY
Iterations	58
Try Max	20
Non-metric R ²	0.9860

7 Site Scores

Table 2: NMDS Site Scores (coordinates in ordination space)

Sample	NMDS1	NMDS2
1	-0.8405	-0.7159
2	-0.5049	-0.4089
3	-0.0827	-0.4367
4	-0.1156	-0.5222
5	-0.6265	-0.0867
6	-0.5427	0.1132
7	-0.5403	-0.0582
8	0.2812	-0.1668
9	0.1106	-0.4426
10	-0.5170	0.0274
11	-0.3383	0.3508
12	0.4425	-0.3641
13	0.4186	-0.5833
14	0.9433	0.4761
15	0.8960	0.2224
16	1.0811	-0.1797
17	-0.8600	0.9871
18	-0.1772	0.5234
19	-0.0701	1.0121
20	1.0424	0.2527

8 Species Scores

Table 3: NMDS Species Scores (top 30 by distance from origin)

Species	NMDS1	NMDS2	Importance
Airaprae	-0.5283	1.6798	1.7609
Empenigr	-0.0884	1.6963	1.6986
Hyporadi	-0.4158	1.4460	1.5046
Callicusp	1.4212	0.3838	1.4721
Comapalu	1.2893	0.6084	1.4257
Eleopalu	1.2451	0.1615	1.2555
Ranuflam	1.1436	0.0996	1.1480
Chenalbu	0.5919	-0.9220	1.0956
Salirepe	0.5748	0.9111	1.0773
Anthodor	-0.7203	0.6591	0.9763
Juncarti	0.9115	-0.0831	0.9152
Cirsarve	-0.1518	-0.8217	0.8356
Achimill	-0.8228	0.0433	0.8239
Elymrepe	-0.4201	-0.6802	0.7995
Trifprat	-0.7715	0.0856	0.7763
Planlanc	-0.7065	0.3206	0.7758
Agrostol	0.7110	-0.2892	0.7675
Vicilath	-0.4679	0.5492	0.7215
Alopgeni	0.3910	-0.5859	0.7044

Table 3: NMDS Species Scores (top 30 by distance from origin) (*continued*)

Species	NMDS1	NMDS2	Importance
Bromhord	-0.6190	-0.3348	0.7037
Juncbufo	0.2648	-0.6076	0.6628
Lolipere	-0.5120	-0.2481	0.5689
Bellpere	-0.4784	-0.2445	0.5372
Rumeacet	-0.5248	-0.1053	0.5352
Poatriv	-0.1591	-0.4784	0.5041
Poaprat	-0.3884	-0.2509	0.4624
Scorautu	-0.1396	0.2500	0.2863
Bracruta	0.1507	0.1898	0.2424
Sagiproc	0.1432	-0.1874	0.2359
Trifrepe	-0.0753	0.0452	0.0878

9 Interpretation

9.1 Ordination Quality

The NMDS analysis achieved a stress value of **0.1183**, which is classified as **B** quality according to Clarke (1993). The configuration should be used with caution. Some distortion of original distances is present.

Consider: (1) increasing dimensions (k), (2) trying different distance metric, or (3) checking for outliers.

The non-metric R^2 value of **0.9860** indicates that 98.6% of the variance in the original dissimilarities is explained by the ordination.

9.2 Convergence

The analysis successfully converged after **58** iterations (maximum: 20 tries). This indicates a stable solution was found.

9.3 Distance Metric

The **BRAY** dissimilarity index was used to quantify differences between samples. This metric is appropriate for abundance data and is robust to differences in sampling effort.

9.4 Biological Interpretation

Samples that are close together in the ordination space have similar community composition, while samples far apart are compositionally dissimilar. The axes in NMDS do not have direct ecological meaning but represent directions of maximum variation in community composition.

9.4.1 Reading the Ordination

- **Proximity:** Sites close to each other have similar species composition
- **Distance:** Greater distances indicate greater compositional dissimilarity
- **Clusters:** Groups of points may indicate distinct community types
- **Gradients:** Smooth transitions may reflect environmental gradients

10 Methods

10.1 Non-metric Multidimensional Scaling (NMDS)

NMDS is an ordination technique that attempts to represent the dissimilarity between samples in a low-dimensional space while preserving the rank order of dissimilarities (Kruskal, 1964). Unlike metric ordination methods (PCA, CA), NMDS:

1. **Does not assume linear relationships** between variables
2. **Preserves rank-order** of dissimilarities rather than exact distances
3. **Is robust to non-normal distributions** and zeros in the data
4. **Works with any dissimilarity measure**

10.1.1 Analysis Parameters

- **Distance Metric:** BRAY dissimilarity
- **Dimensions:** 2
- **Number of sites:** 20
- **Number of species:** 30
- **Random starts:** 20
- **Convergence criterion:** Stress reduction

10.1.2 Algorithm

The NMDS algorithm iteratively adjusts sample positions to minimize stress, which measures the departure from monotonicity in the relationship between observed dissimilarities and ordination distances. The algorithm:

1. Starts with a random (or PCA-based) configuration
2. Calculates distances between points in ordination space
3. Compares these with observed dissimilarities
4. Moves points to reduce stress
5. Repeats until convergence or maximum iterations reached

10.2 Stress Interpretation Guidelines

The quality of NMDS ordination is assessed using the stress value (Clarke, 1993):

- **Stress < 0.05:** Excellent representation with no prospect of misinterpretation
- **Stress < 0.10:** Good ordination with no real risk of drawing false inferences
- **Stress < 0.20:** Potentially useful, but interpret with caution
- **Stress 0.20:** Poor representation; points are close to being arbitrarily placed

10.3 Goodness of Fit

The goodness of fit for each site indicates how well that sample is represented in the ordination. Values close to 1 indicate good representation, while lower values suggest the sample's position may not accurately reflect its true dissimilarity to other samples.

10.4 Shepard Diagram

The Shepard plot displays the relationship between:

- **X-axis:** Observed dissimilarities (from the original distance matrix)
- **Y-axis:** Ordination distances (Euclidean distances in the reduced space)
- **Step line:** Monotonic relationship that NMDS attempts to achieve
- **Scatter:** Actual fit of each pairwise comparison

Points closer to the step line indicate better representation of those particular sample pairs.

11 Reproducibility

11.1 Analysis Configuration

This section documents all parameters and settings used in the analysis to ensure full reproducibility.

11.1.1 Dataset Information

- **Dataset Name:** Dune Meadow Vegetation
- **Number of Sites/Samples:** 20
- **Number of Species/Variables:** 30
- **Analysis Date:** October 25, 2025
- **Analysis Time:** 18:10:29

11.1.2 NMDS Parameters

The following parameters were used for the NMDS ordination:

Table 4: Complete NMDS Analysis Configuration

Parameter	Value	Description
Distance/Dissimilarity Metric	BRAY	Dissimilarity measure used to quantify differences between samples
Number of Dimensions (k)	2	Number of ordination axes/dimensions
Maximum Random Starts	20	Number of random starting configurations attempted
Autotransformation	No	Whether data were automatically transformed (e.g., Wisconsin double standardization)
Permutations (stress test)	999	Number of permutations for statistical testing
Convergence Achieved	Yes	Whether the iterative algorithm reached a stable solution
Final Iterations	58	Number of iterations before convergence or termination
Final Stress Value	0.1183	Final stress value indicating quality of ordination

11.1.3 Software Environment

Table 5: Software Environment

Component	Version
Analysis Platform	Ördin
Platform Version	v3.0
R Version	4.5.1
vegan Package	2.7.2
Operating System	Windows

11.1.4 Reproducing This Analysis

To reproduce this exact analysis:

1. **Load your data** into R with the same structure:
 - Rows = Sites/Samples (20 sites)
 - Columns = Species/Variables (30 species)

2. Install required packages:

```
install.packages("vegan")  
library(vegan)
```

3. Run NMDS with identical parameters:

```
nmds_result <- metaMDS(  
  comm = your_data,           # Your community matrix  
  distance = "bray",  
  k = 2,  
  trymax = 20,  
  autotransform = FALSE,  
  trace = FALSE  
)
```

4. Verify convergence:

```
nmds_result$converged # Should be TRUE  
nmds_result$stress    # Should be 0.1183
```

Note: Due to random starting configurations, exact site coordinates may differ slightly between runs, but stress values and overall patterns should be nearly identical.

12 References

- Clarke, K.R. (1993). Non-parametric multivariate analyses of changes in community structure. *Australian Journal of Ecology*, 18(1), 117-143. <https://doi.org/10.1111/j.1442-9993.1993.tb00438.x>
- Kruskal, J.B. (1964). Multidimensional scaling by optimizing goodness of fit to a nonmetric hypothesis. *Psychometrika*, 29(1), 1-27. <https://doi.org/10.1007/BF02289565>
- Minchin, P.R. (1987). An evaluation of the relative robustness of techniques for ecological ordination. *Vegetatio*, 69(1-3), 89-107. <https://doi.org/10.1007/BF00038690>
- Oksanen, J., et al. (2024). vegan: Community Ecology Package. R package version 2.7-2. <https://CRAN.R-project.org/package=vegan>

13 Software

This analysis was performed using:

- **Ördin v3.0** - Community Ecology Analysis Platform
- **R** version 4.5.1
- **vegan** package version 2.7.2
- **Generated:** 2025-10-25 18:10:30 +10

Report generated by Ördin v3.0

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